

Predators & Invasive Threats

Interactive Implementation Guide

Detailed implementation guide for building the interactive “Predators & Invasive Threats” section of the Hawaiian Coot conservation website.

1. Core Interactive Layout

The page should be divided into four major areas:

- Interactive predator cards
- Cinematic ecosystem viewer
- Threat metrics dashboard
- Ecosystem comparison carousel

The entire page reacts dynamically depending on the predator selected.

2. Main Ecosystem Viewer

The center panel is the primary storytelling area.

Requirements:

- Large cinematic wetland scene
- Hawaiian Coot visible in ecosystem
- Dynamic lighting changes

- Predator activity overlays
- Nest warning indicators

Suggested States:

- Healthy habitat
- Predator active
- Nest under threat
- Recovery mode

3. Mongoose Interaction

Hover or click the Mongoose card.

Visual Changes:

- Wetlands darken
- Red warning glow intensifies
- Nest icons flash red
- Threat scan overlay appears

Animations:

- Mongoose silhouette moves through grass
- Birds scatter
- Nest disturbance effect
- Thermal predator scan animation

4. Rats & Feral Cats Interaction

Hover or click Rats & Feral Cats.

Visual Changes:

- Night atmosphere appears
- Red glowing eyes appear
- Nest warnings pulse

Animations:

- Shadow movement
- Egg disappearance effects
- Heartbeat warning pulses

5. Dogs Interaction

Hover or click Dogs.

Visual Changes:

- Nesting areas disturbed
- Birds fly away suddenly
- Wetland pathways damaged

Animations:

- Paw print movement
- Disturbance ripple effects
- Sudden flight animations

6. Bullfrogs Interaction

Hover or click Bullfrogs.

Visual Changes:

- Murky swamp tones
- Food-chain imbalance indicators
- Insect activity decreases

Animations:

- Frog ripple effects
- Water vibration pulses
- Competition indicators

7. Night Heron Interaction

IMPORTANT: Night Heron should feel natural rather than dangerous.

Visual Changes:

- Calm moonlit atmosphere
- Balanced ecosystem glow
- Gentle wetland reflections

Animations:

- Elegant bird flight

- Smooth water ripples
- Calm environmental movement

8. Ecosystem Health Dashboard

Add a live ecosystem status dashboard.

Suggested Metrics:

- Habitat Health
- Nesting Success
- Water Quality
- Food Availability
- Population Stability

Dynamic color states:

- Green = healthy
- Yellow = stressed
- Red = critical

9. Image Asset Requirements

Recommended Major Images:

- Healthy Wetland
- Threatened Wetland
- Mongoose Impact Scene

- Rats & Cats Impact Scene
- Dogs Disturbance Scene
- Bullfrog Competition Scene
- Night Heron Natural Predation Scene
- Extreme Habitat Collapse Scene

Recommended Overlay Assets:

- Threat scans
- Nest warnings
- Predator silhouettes
- Ripple effects

10. Recommended Technologies

Recommended Libraries:

- React
- Framer Motion
- GSAP
- CSS keyframes

Suggested Animation Types:

- Ripple shaders
- Glow pulses
- Dynamic lighting
- Particle systems
- Smooth crossfade transitions

11. Interaction Trigger Table

USER ACTION	VISUAL CHANGE	ANIMATION	PRIMARY IMPACT
Select Mongoose	Wetlands darken	Threat scan + nest warning	Chick Survival decreases
Select Rats & Cats	Night atmosphere	Shadow movement + pulse	Nest predation increases
Select Dogs	Birds scatter	Paw print disturbance	Nesting disturbance rises
Select Bullfrogs	Murky wetlands	Ripple vibration effects	Food competition rises
Select Night Heron	Moonlit ecosystem	Smooth bird flight	Natural ecological balance
Hover Threat Metrics	Indicators expand	Glow pulse animation	Educational emphasis

12. Final Design Philosophy

The Predators & Invasive Threats section should feel like a cinematic ecosystem threat simulator. Every interaction should visually reinforce how invasive predators disrupt Hawaiian wetland ecosystems and threaten endangered Hawaiian Coot populations.